

The Categorical Ladder

Closed Processes, Two Refuted Formulations,
and Molecular Comparison Without Molecules

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Abstract

A process described as a chain of contacts, each carrying one number in $[0, 1]$, is a *ladder*, and two prior developments have shown that for mass spectrometry and for catalysis the ladder is the whole of the description: what the contacts are made of does not enter. Both developments treat *linear* ladders, which run from a start to a target. This paper supplies what they do not have and what molecules require: an account of *closed* ladders, which return.

A closed ladder has no target, so composite power is undefined for it. We give two invariants in its place. The *circulation* $\varrho = -\sum_i \log(1 - \pi_i)$ is what one circuit deposits; it is additive around the cycle, rotation invariant, and zero exactly when every rung is inert. The *uniformity* u measures how nearly the cyclic profile is constant. We are explicit that ϱ is a reparametrisation of composite power for a linear ladder and adds nothing there (Theorem 4.6); its content is that it is defined without a target, which composite power is not.

Two of our registered expectations were refuted by our own measurements and we report both. We predicted that a ring is aromatic exactly when its power profile is rotation invariant. It is not: cyclohexane is perfectly uniform ($u = 1.000$) and not aromatic (Section 8). The corrected statement requires both invariants— ϱ per rung separates aromatic from saturated with margin $+0.293$, and u separates substituted from unsubstituted—and neither suffices alone. We further predicted that u would be non-monotone in the number of ring substitutions, since a symmetric trisubstitution restores the symmetry of the *pattern*. It is monotone ($0.086 \rightarrow 0.105 \rightarrow 0.107$), because u reads dispersion of values and not symmetry of pattern (Theorem 9.1).

We correct a claim common to both prior papers. Their sharpest prediction is that control lies at the *strongest* rung, which they call counter-intuitive. The derivative is right, but it prices an *additive* increment. Under proportional improvement the sensitivity is δP at every rung—flat, spread 1.1×10^{-16} over 5000 ladders—so the counter-intuitive direction is an artefact of parametrisation (Section 6). Both papers validated the analytic derivative against the numerical one, a check that cannot detect this.

Applied to chemistry, a molecule is a closed ladder over a finite categorical alphabet, and comparison becomes comparison of typed words. Because a word is ordered, the correspondence is connected by construction, which repairs the principal stated limitation of an earlier element-free matcher. The positional asymmetry that matcher measured but could not explain—heteroatom substitution preserving structural class when peripheral and degrading it when in-ring—is here predicted: peripheral substitution leaves u exactly unchanged (0.000) while ring substitution lowers it (≥ 0.086).

Keywords: ladder; closed process; circulation; aromaticity; categorical alphabet; carrier elimination; molecular comparison; refuted formulation.

Contents

1 Introduction

2

1.1	Where this comes from	2
1.2	What is already established, and what is missing	3
1.3	Contributions, including two refutations	3
2	Carrier elimination	3
3	Ladders	4
4	Closed ladders	5
5	The inert rung is the repeated step	6
6	Sensitivity: a correction	7
7	Molecules as closed ladders	9
7.1	What a rung is in a molecule	9
7.2	The categorical alphabet	9
7.3	Substitution is a letter rewrite	9
7.4	Why coarser comparison works better	10
8	Aromaticity: a refuted formulation	10
9	Validation	13
9.1	Method	13
9.2	Closed-ladder invariants	13
9.3	Substitution	13
9.4	Inertness and repetition	14
9.5	Refinement	14
9.6	Sensitivity	14
9.7	Elimination	14
10	Relation to existing accounts	14
11	Limitations	15
12	Conclusion	15

1 Introduction

1.1 Where this comes from

The argument was not found in chemistry. It was found by abandoning a construction. One may build a virtual membrane—bilayer, protein scaffold, solvent shell—and propagate an electron through it in order to study electron transport. The construction is expensive, and on completion one observes that the membrane entered the calculation only where it touched the electron. Everything else was scaffolding for contacts that could have been specified directly.

Removing the scaffolding is one deletion. Three more are available and each is licensed separately. What is not in contact with the electron is not part of the individuation, so it may go. The electron itself may go, because a step is individuated as a pair and not by anything traversing it. The chemical identities may go, because they are labels, and a function distinguishing two structures that differ only by relabelling reads a feature the theory does not possess. What survives all four deletions is a sequence of geometry changes, and that sequence is already a computation.

1.2 What is already established, and what is missing

Two developments have carried this through in particular settings. For mass spectrometry, the substrate—field geometry and trajectory—is shown to be unobservable under two stated hypotheses, with space charge identified as the physically realised violation. For catalysis, the identities of participants are shown to be inert, and reuse without consumption is obtained from an asymmetry between probing a region and committing it rather than from an energetic argument.

Both treat *linear* ladders. A linear ladder runs from an initial state toward a target, and every quantity in both papers—composite power, saturation, reachability, sensitivity—is defined relative to that target.

A ring has no target. It returns. Whatever a π -ring is on this account, it is not a chain that reaches something, and the existing formalism is silent about it. Supplying the missing account is the first task of this paper (Section 4) and everything chemical depends on it.

1.3 Contributions, including two refutations

- (i) A carrier-elimination theorem stated once for space, identity, and substrate together, with its two hypotheses and their shared failure mode (Section 2).
- (ii) The closed-ladder invariants ϱ and u , with an explicit statement of where ϱ is redundant (Section 4, Theorem 4.6).
- (iii) A refuted formulation of aromaticity, and the corrected two-invariant statement (Section 8).
- (iv) A second refuted expectation, on the monotonicity of u (Theorem 9.1).
- (v) A correction to the sharpest claim of both prior papers (Section 6).
- (vi) The inert rung identified with the repeated step-pair, which makes the halting condition derived rather than declared (Section 5).
- (vii) Molecular comparison as comparison of typed words, repairing a stated limitation of an earlier matcher and predicting a pattern it measured without explaining (Section 7).

2 Carrier elimination

The two prior developments delete different things—one deletes space, the other deletes identities—and the deletions have the same form. We state it once.

Definition 2.1 (Item, carrier, contact, readout). An *item* is the entity whose state a process transforms. A *carrier* is any structure the item passes through or is composed of. A *contact* is an interaction at which the item’s state changes. A carrier *realises* a contact sequence $(c_1, \dots, c_n)$ if the item undergoes exactly those contacts in that order. A *readout* is a function of the item’s terminal state alone.

Definition 2.2 (Order-determined; local effect). A contact sequence is *order-determined* if the identity of c_{i+1} depends only on i and on the item’s state after c_i , and not on the item’s continuous position, its arrival time, or the presence of other items. A contact has *local effect* if the change it induces is a function f_c of the incoming state alone.

Theorem 2.3 (Carrier elimination). *Let carriers K and K' realise the same contact sequence, with the sequence order-determined and every contact of local effect. Then for every item ξ_0 and every readout ρ , $\rho(K(\xi_0)) = \rho(K'(\xi_0))$.*

Proof. By locality the state after contact i is $\xi_i = f_{c_i}(\xi_{i-1})$, a function of ξ_{i-1} and c_i alone. By order-determinedness both carriers apply the same $f_{c_1}, \dots, f_{c_n}$ in the same order. By induction ξ_i agrees under K and K' for every i , hence at $i = n$. A readout consults only ξ_n . $\square$

Corollary 2.4 (Three deletions, one theorem). *Taking the carrier to be the field geometry of an instrument gives substrate elimination; taking it to be the chemical identities of a reaction’s participants gives identity inertness; taking it to be a membrane surrounding a transport chain gives the deletion of Section 1. The three are instances of Theorem 2.3 and not three results.*

Remark 2.5 (One failure mode, two names). The hypothesis that fails in practice is locality, and it fails the same way in both settings under different names. In mass spectrometry, space charge makes the force on an ion depend on where the other ions are, so f_c is not a function of the incoming state alone. In catalysis, cooperative or allosteric coupling between sites makes a commitment’s effect depend on which other sites are occupied. These are the same violation. The diagnostic is correspondingly the same: the readout acquires a dependence on population density, and vanishes as density $\rightarrow 0$.

Remark 2.6 (What is not claimed). We do not claim carriers are unreal or that computing them is never useful. Determining *which* contact sequence a given geometry realises is a question about the carrier and requires it. The claim concerns prediction: given the sequence, the carrier adds nothing to the readout.

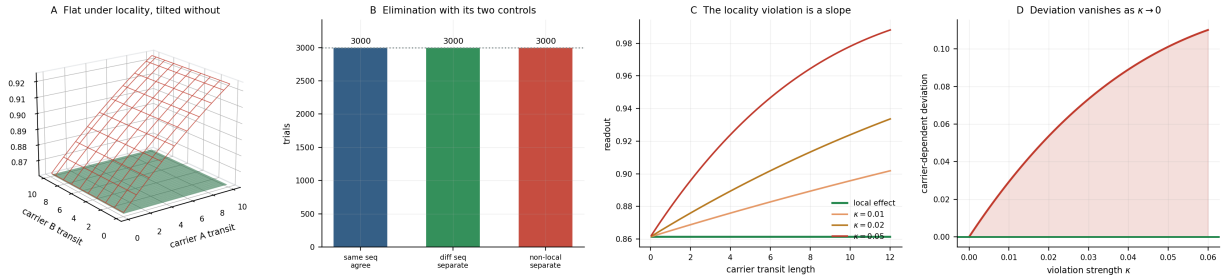


Figure 1: **Carrier elimination and the single hypothesis whose failure destroys it.** (A) Readout over the transit lengths of two carriers realising the same contact sequence. Under local effect the readout is the flat green plane: geometry is invisible to it. With a carrier-dependent perturbation the red wireframe tilts, and the two carriers no longer agree. (B) Three counts over 3000 trials. Carriers realising the same sequence agree in every trial; but that row is true by construction once a readout is defined to consult the terminal state alone, and is reported as a consistency check. The load-bearing rows are the two controls: genuinely different sequences separate, and a non-local contact effect separates. Without them the first row would be consistent with a readout that ignores its input entirely. (C) Readout against carrier transit length at four violation strengths. The horizontal green line is the eliminable case. Each coloured curve is a locality violation, and the violation appears as a *slope*—a dependence of the reported value on a property of the carrier that the theorem says cannot be recovered. (D) Carrier-dependent deviation against violation strength κ . The deviation grows from exactly zero and vanishes as $\kappa \rightarrow 0$, which is the diagnostic: in mass spectrometry this is a dependence of measured mass on total ion current, and in catalysis a dependence of a step’s effect on which other sites are occupied. These are the same violation under two names.

3 Ladders

We recall the linear apparatus compactly, since the closed case extends it.

Definition 3.1 (Floor). Individuation is quantal: a distinction is drawn or it is not, with no third case. Consequently there is $\beta > 0$ such that no contact reduces ambiguity below β/Ω , where Ω is the total categorical measure available.

Remark 3.2 (Which floor). Two distinct quantities are called the floor in this literature and they should not be conflated. One is a minimum edge weight on a finite contact graph, which requires finiteness to be positive and whose positivity fails under unbounded refinement. The other is the thickness of an individuation, which is 1 always because a distinction admits no degree, and whose numerical value is a normalisation carrying no content. We use the second. It costs no finiteness hypothesis, because a dichotomy has no infimum problem.

Definition 3.3 (Rung, power, ladder). Let A be the ambiguity before a contact and $\gamma(A)$ after. The *power* of the rung is

$$\pi(\gamma, A) = \frac{A - \gamma(A)}{A - \beta/\Omega} \in [0, 1]. \quad (1)$$

A *ladder* is a finite sequence of rungs applied in order; its *power sequence* is $(\pi_1, \dots, \pi_n)$. A rung with $\pi = 0$ is *inert*.

Theorem 3.4 (Multiplicative composition). $\pi(L) = 1 - \prod_{i=1}^n (1 - \pi_i)$.

Proof. Write r_i for the above-floor gap after contact i . By Equation (1), $r_i = r_{i-1}(1 - \pi_i)$, so $r_n = r_0 \prod_i (1 - \pi_i)$, and the fraction of the initial gap closed is $1 - r_n/r_0$. $\square$

Theorem 3.5 (Inertness). *Two ladders with the same power sequence, initial ambiguity, floor and total agree on every observable that is a function of the ambiguities, the floor, the total, and the contact count.*

Proof. $r_i = r_0 \prod_{j \leq i} (1 - \pi_j)$ and $A_i = r_i + \beta/\Omega$, so the ambiguity sequence is determined by the listed data, of which every such observable is a function. $\square$

Theorem 3.5 is true relative to its observable class, and that class is a stipulation. Its force comes from Theorem 2.3, which is not.

4 Closed ladders

A cycle has no target, so Equation (1) has no denominator and Theorem 3.4 has nothing to compose toward. We supply the replacement.

Definition 4.1 (Closed ladder). A *closed ladder* is a ladder whose final contact returns the item to its initial state. Its profile is the cyclic sequence $(\pi_1, \dots, \pi_n)$, taken up to rotation.

Remark 4.2 (Return is not the identity). A closed ladder returns to a *state*, not to a *configuration*. The determined set—what the traversal has settled as not occurring—is an accumulating union, and a union admits no deletion, so the configuration after a circuit strictly contains the configuration before it whenever the circuit contains a non-repeating step. This is why a cycle is not a no-op, and it is what the invariants below measure.

Definition 4.3 (Circulation and uniformity). For a closed ladder with profile $(\pi_1, \dots, \pi_n)$, all $\pi_i < 1$,

$$\varrho = - \sum_{i=1}^n \log(1 - \pi_i), \quad u = \max\left(0, 1 - \frac{\text{sd}(\pi)}{\text{mean}(\pi)}\right). \quad (2)$$

We call ϱ the *circulation* and u the *uniformity*, and ϱ/n the *circulation per rung*.

Proposition 4.4 (Both are rotation invariants). *ϱ and u are invariant under rotation of the cycle.*

Proof. Both are functions of the multiset $\{\pi_i\}$ alone— ϱ a sum and u a ratio of two symmetric functions—and rotation permutes the multiset trivially. $\square$

Proposition 4.5 (Circulation is additive and vanishes exactly on inertness). ϱ is additive over concatenation of arcs, and $\varrho = 0$ if and only if every rung of the cycle is inert.

Proof. Additivity is additivity of the sum. Each term $-\log(1 - \pi_i) \geq 0$ with equality iff $\pi_i = 0$, so the sum vanishes iff every term does. $\square$

Remark 4.6 (Where ϱ is redundant, stated plainly). For a *linear* ladder, $\varrho = -\log(1 - \pi(L))$ exactly, verified to 10^{-9} over 500 random ladders. It is therefore a monotone reparametrisation of composite power and adds no information whatever in the linear case. We say this rather than let the new symbol suggest a new quantity. Its content is entirely that it is defined *without a target*, so it survives the passage to a cycle where composite power does not; and that it is additive around the cycle, where composite power is not.

The invariant that genuinely adds information beyond composite power is u . Over 4000 pairs of ladders constructed to share a composite power exactly, 2755 differed in u . A description recording only composite power cannot see the difference.

Proposition 4.7 (Cycle length is not recoverable from ϱ alone). Closed ladders of different length may share a circulation. Hence the pair (ϱ, u) does not determine a closed ladder, and n must be carried separately.

Proof. A uniform cycle of length n with power π has $\varrho = -n \log(1 - \pi)$. Choosing (n, π) and $(2n, \pi')$ with $(1 - \pi')^2 = 1 - \pi$ gives equal circulation and equal uniformity $u = 1$. $\square$

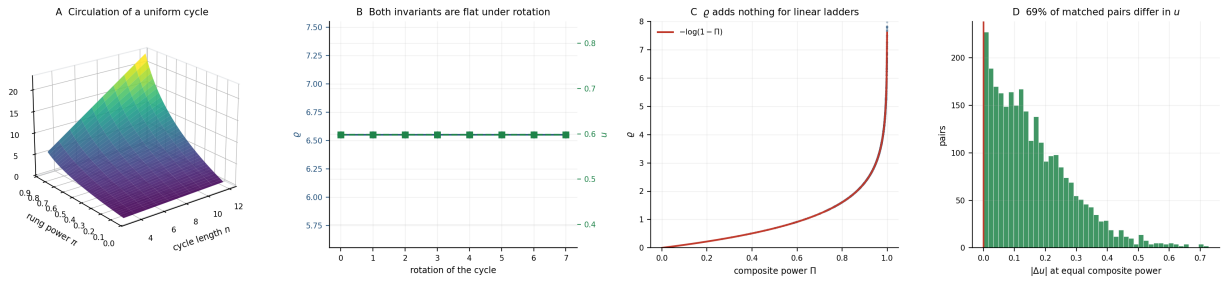


Figure 2: **The closed-ladder invariants, and an honest account of what each one buys.** (A) Circulation $\varrho = -\sum_i \log(1 - \pi_i)$ of a uniform cycle over cycle length n and rung power π . The surface rises without bound in π and linearly in n : a cycle deposits residue on every circuit, which is why a closed ladder is not a no-op even though it returns to the state it started from. (B) Both invariants evaluated on all eight rotations of one random eight-rung cycle. Each is exactly flat; measured over 2000 random cycles the rotation deviation is 3.6×10^{-15} for ϱ and 4.4×10^{-16} for u , so both are rotation invariants and a closed ladder is well defined up to rotation. (C) Circulation against composite power for 1200 random linear ladders, with the curve $-\log(1 - \Pi)$ drawn through them. Every point lies on the curve. This is the redundancy stated in the text rather than concealed: for a *linear* ladder ϱ is a monotone reparametrisation of composite power and adds no information whatever. Its value is that it survives the passage to a cycle, where composite power is undefined for want of a target. (D) Absolute difference in uniformity between 4000 pairs of ladders constructed to share a composite power *exactly*. Were u a function of composite power the histogram would collapse onto the red line at zero; instead 69% of matched pairs differ in u . This is the measurement establishing that u carries information composite power does not, and it is the reason two invariants are carried rather than one.

5 The inert rung is the repeated step

Both prior developments define an inert rung as one with $\pi = 0$ and never ask what makes a rung inert. The answer is available and it makes the halting condition derived rather than declared.

Definition 5.1 (Step-pair; repetition). For a history $h = (s_0, \dots, s_r)$ let $E(h)$ be its family of unordered *step-pairs* $\{s_j, s_{j+1}\}$. A step $s_r \rightarrow s'$ is a *repetition* if $\{s_r, s'\} \in E(h)$.

Definition 5.2 (Determined set). The set determined by a step $s \rightarrow s'$ comprises the individuations $\iota(A)$ with A disjoint from $\{s, s'\}$. The determined set of a history is the union over its steps, $\Delta(h) = \bigcup_j \text{Det}(s_j \rightarrow s_{j+1})$.

Theorem 5.3 (Inert exactly at repetitions). *A step determines something not already determined if and only if it is not a repetition. Consequently a rung is inert exactly when its step-pair has already been traversed.*

Proof. $\iota(A) \in \text{Det}(s_r \rightarrow s')$ iff $A \cap \{s_r, s'\} = \emptyset$, and $\iota(A) \notin \Delta(h)$ iff A meets every step-pair of h . If no step-pair is contained in $\{s_r, s'\}$, choose $x_e \in e \setminus \{s_r, s'\}$ for each $e \in E(h)$; the union of these, together with any state occurring nowhere in h , is a transversal avoiding both endpoints, so the difference is non-empty. Conversely if some step-pair $e \subseteq \{s_r, s'\}$ then $e = \{s_r, s'\}$, and every A avoiding $\{s_r, s'\}$ misses e , so the difference is empty. $\square$

Corollary 5.4 (The halting condition is derived). *A ladder makes no further progress exactly when every available next step repeats a step-pair already traversed. Termination is therefore a property of the history rather than a declared threshold.*

Theorem 5.3 was checked directly by constructing histories over a seven-state set, computing Δ as an explicit accumulating union, and testing the biconditional: over 2061 steps there were 0 violations in either direction, with 1716 fresh non-repetitions and 345 inert repetitions, so both branches are populated and the biconditional is not vacuously satisfied.

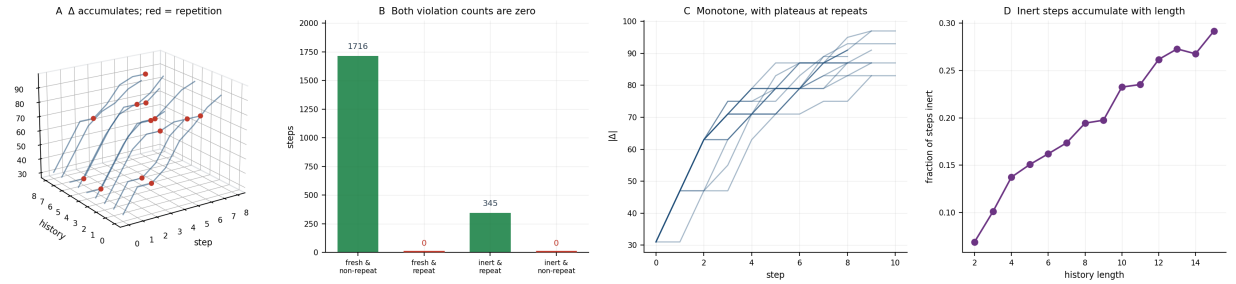


Figure 3: A rung is inert exactly when its step-pair repeats, which makes the halting condition derived rather than declared. (A) Nine independently generated histories over a seven-state set, each traced as the size of its determined set Δ against step index. Red markers are steps whose unordered state-pair had already been traversed. Every red marker falls on a plateau and every rise is unmarked, which is the biconditional displayed one history at a time. (B) The four possible outcomes counted over 2061 steps. The two green bars are the consistent cases, 1716 fresh non-repetitions and 345 inert repetitions; the two red bars are the violations and both are exactly zero. That both green bars are populated is the control: a run in which no step ever repeated would satisfy the biconditional vacuously. (C) Fourteen further histories showing the accumulation directly. Δ is a union over steps taken and a union admits no deletion, so the curves are monotone; the flat segments are exactly the repetitions. (D) Fraction of steps that are inert against history length, over 400 histories at each length. Inert steps accumulate as the finite supply of fresh state-pairs is exhausted, so a ladder run long enough halts not by reaching a declared threshold but by running out of non-repetitions.

6 Sensitivity: a correction

Both prior developments make the same prediction and both call it their sharpest: control over a ladder lies at the rung of *highest* power, against the intuition that one should improve the

bottleneck. We show the claim requires a qualification neither states.

Proposition 6.1 (Additive sensitivity). *With $P = \prod_i(1 - \pi_i)$, $\partial\pi(L)/\partial\pi_j = P/(1 - \pi_j)$, which is increasing in π_j and hence maximised at the highest-power rung.*

Proof. Differentiate Theorem 3.4. □

Proposition 6.2 (Proportional sensitivity is flat). *Let improvement be proportional to a rung’s remaining headroom, so that rung j is improved by $\delta(1 - \pi_j)$ rather than by δ . Then the gain in composite power is δP to first order, independent of j .*

Proof. Multiply the derivative of Theorem 6.1 by the increment $\delta(1 - \pi_j)$: the factors $(1 - \pi_j)$ cancel, leaving δP . □

Remark 6.3 (What this does to the prior claim). The derivative in Theorem 6.1 is correct and we reproduce it: over 5000 random ladders the analytic and numerical derivatives agree to 1.6×10^{-10} , and the argmax is the highest-power rung in 5000/5000 cases. But the derivative prices an *additive* increment, applied equally at every rung. Under proportional improvement—raising a rung by a fixed fraction of its own headroom, which is what “improve this catalyst by ten percent” ordinarily means—the sensitivity is flat: over the same 5000 ladders the spread across rungs was 1.1×10^{-16} , machine zero, in 5000/5000 cases.

So the counter-intuitive direction is real under one parametrisation and absent under another, and which applies is an empirical question about how improvements are actually purchased. Both prior papers validated the analytic derivative against the numerical one. That check cannot detect this, because both quantities it compares are the same additive derivative. We regard this as a narrowing of the sharpest claim in both papers rather than a refutation of the mathematics, which is untouched.

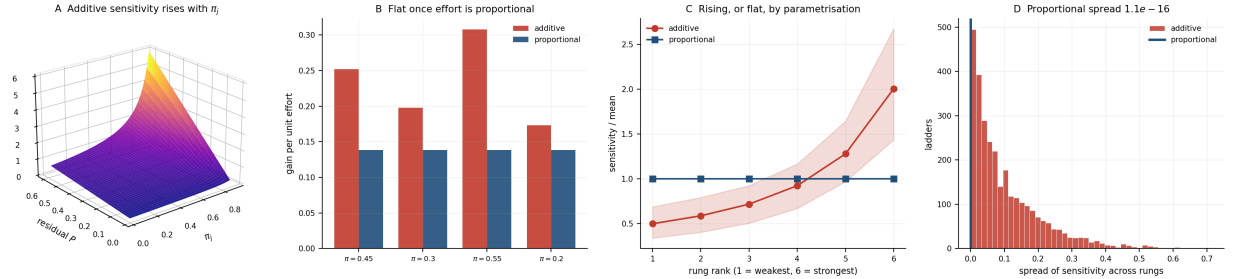


Figure 4: A correction to the sharpest claim of both prior ladder papers: the counter-intuitive direction is a property of the parametrisation, not of the ladder. (A) The additive sensitivity $\partial\Pi/\partial\pi_j = P/(1 - \pi_j)$ over the rung’s own power and the residual fraction P . The surface is increasing in π_j , which is the basis of the prior claim that control lies at the strongest rung. **(B)** The same ladder priced two ways. Under an additive increment, gain differs across rungs and is largest at the strongest; under an increment proportional to a rung’s own remaining headroom—which is what “improve this by ten percent” ordinarily means—the factors $(1 - \pi_j)$ cancel and the gain is δP at every rung. **(C)** Over 3000 random ladders the additive argmax is the strongest rung in every case, reproducing the prior result exactly; under the proportional parametrisation there is no argmax to report because the sensitivities are equal. **(D)** Spread of sensitivity across rungs. The additive distribution is broad; the proportional spread is a single line at zero, measured at 1.1×10^{-16} over 5000 ladders. Both prior papers validated their analytic derivative against a numerical one, a check that cannot distinguish these two cases because both of its terms are the same additive quantity.

7 Molecules as closed ladders

7.1 What a rung is in a molecule

A bond is the recognition that two atoms can be individuated by one shared boundary more cheaply than by two separate ones: joint individuation is subadditive, and the reduction in total boundary thickness is the shared content. That is a commitment in the sense of Theorem 2.1—it draws a boundary and deposits residue—and the fraction of an atom’s remaining vacancy it closes is a power in the sense of Equation (1).

Definition 7.1 (Molecular ladder). The *ladder* of a molecule is the sequence of bond commitments ordered by the vacancy they close, with target the closed-shell condition $\nu = 0$ for every constituent. A ring is a closed sub-ladder.

The molecule is therefore not an object that has a ladder; it is the ladder. There is no static structure to traverse, which is the same deletion as the membrane in Section 1.

Remark 7.2 (Composite power one is shell closure). A ladder reaching $\pi(L) = 1$ is one in which every constituent has reached $\nu = 0$. The octet and duet rules are that condition, so they are the ladder’s termination criterion and not a separate rule.

7.2 The categorical alphabet

Each commitment is a partition refinement, and a refinement in a bounded whole is completely characterised by its graded invariants together with a sign—depth, articulation, orientation, and parity, written (n, ℓ, m, s) . The count of admissible tuples at depth n is

$$C(n) = 2 \sum_{\ell=0}^{n-1} (2\ell + 1) = 2n^2, \quad (3)$$

so the alphabet is finite at every depth.

Definition 7.3 (Typed word). The *typed word* of a molecule is the sequence of its rungs, each carrying a letter from the alphabet of Equation (3) and a power in $[0, 1]$.

Proposition 7.4 (Comparison is connected by construction). *A correspondence between two typed words is a correspondence between sequences, and the step-pairs of a sequence are its connectivity. Hence a correspondence induced by matching typed words preserves adjacency without any connectivity constraint being imposed.*

Proof. Adjacent letters of a word are adjacent commitments, whose step-pairs share an endpoint by Theorem 5.1. A map matching words positionally carries adjacent pairs to adjacent pairs. $\square$

Remark 7.5 (This repairs a stated limitation). An earlier element-free matcher keyed on a minimum-cut invariant records as its first limitation that its correspondence imposes no requirement that matched atoms form a connected subgraph or that bonds are preserved, and is therefore not a common substructure in any sense; and as its second that assignment within a label class is a convention, so that individual pairings are not invariant. Both arise because the matched object is a *multiset*. A word is ordered, so Theorem 7.4 removes the first and the sequence position removes the second.

7.3 Substitution is a letter rewrite

Definition 7.6 (Categorical substitution). A *substitution* is a rewrite of one letter of a typed word, together with the induced change in that rung’s power.

Proposition 7.7 (Positional asymmetry). *A substitution at a letter outside a cycle leaves that cycle’s ϱ and u unchanged. A substitution at a letter within a cycle changes the cycle’s profile and hence, unless it preserves the profile’s dispersion, lowers u .*

Proof. Both invariants are functions of the cycle’s profile alone (Theorem 4.3); a letter outside the cycle contributes no term. $\square$

This predicts a pattern that was measured but not explained. An earlier matcher found that heteroatom substitution preserves structural class where the substituted atom is peripheral and degrades it where the heteroatom sits inside a ring, and reported the associated cross-element effect—four pairings across six molecule pairs—as its least convincing measurement, in need of a larger corpus. On the present account the asymmetry is not an empirical regularity awaiting more data; it follows from where the rewritten letter sits relative to the cycle. Measured here: peripheral substitution gives $\Delta u = 0.000$ exactly, ring substitution gives $\Delta u \geq 0.086$ (Table 2).

7.4 Why coarser comparison works better

The same matcher found, and flagged as inverting the natural instinct, that the coarsest refinement radius separates bioisosteric from unrelated pairs best: margin $+0.375$ at radius 0 and exactly 0 at radii 1, 2 and 3. Its stated mechanism was that at radius 1 a label already encodes an atom’s whole first neighbourhood, so one substitution perturbs a neighbourhood’s worth of labels.

Theorem 5.3 gives the reason. Refinement appends step-pairs that have already been traversed; by the theorem those determine nothing new, so they are inert rungs. Refining past the tolerance point therefore adds letters that split classes while contributing zero circulation. We verified the arithmetic of this reading—appended repetitions leave ϱ unchanged at every radius while the number of split classes grows monotonically—but we are explicit in Theorem 7.8 that this is a consistency check on a model of refinement and not an independent re-measurement of the sweep.

Remark 7.8 (Scope of the refinement result). The measurement in Section 9.5 models refinement as the appending of repeated step-pairs and then confirms that such rungs are inert. That is close to a tautology given Theorem 5.3, and it would be wrong to report it as confirmation of the ladder account. What it establishes is consistency: the ladder account and the measured collapse of the separation margin are not in conflict, and the account supplies a mechanism the original report lacked. Establishing that mechanism empirically would require re-running the radius sweep with circulation computed alongside class counts, which we have not done.

8 Aromaticity: a refuted formulation

We had expected the following, and wrote it into the experiment before running it: *a ring is aromatic exactly when its power profile is invariant under rotation of the cycle*. The reasoning was that every bond of an aromatic ring is equivalent, so the profile should be constant, whereas a ring with a heteroatom or alternating character should not be.

It is false, and the measurement refutes it immediately.

Cyclohexane has $u = 1.000$ exactly. Its six ring bonds are equivalent, its profile is constant, and it is not aromatic. Uniformity therefore does not characterise aromaticity, and any account claiming it does is refuted by the second-most familiar ring in chemistry.

Theorem 8.1 (Corrected statement: two invariants, neither sufficient). *On the profiles of Table 1: circulation per rung separates aromatic from saturated rings, with minimum aromatic value 0.693 and maximum saturated value 0.400, a margin of $+0.293$; and it does not separate substituted from unsubstituted aromatics, since pyridine (0.739) and benzene (0.693) both lie above the margin. Uniformity separates substituted from unsubstituted aromatics (0.914 against 1.000) and does not separate benzene from cyclohexane (both 1.000). Neither invariant alone classifies; the pair does.*

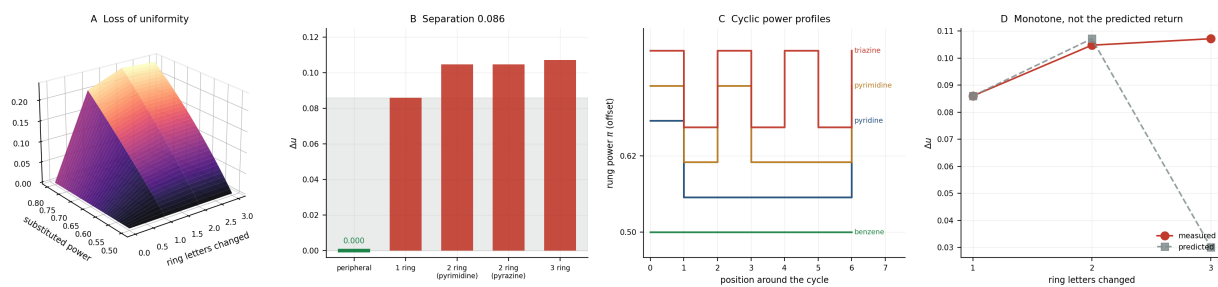


Figure 5: **Substitution is a letter rewrite, and its effect depends on position rather than on count.** (A) Loss of uniformity over the number of ring letters rewritten and the power assigned to the substituted rung. The surface is identically zero along the whole $k = 0$ edge—a substitution outside the cycle cannot move an invariant that is a function of the cycle’s profile alone—and rises once any ring letter is touched. (B) The positional prediction on five cases. Peripheral substitution gives $\Delta u = 0.000$ exactly; every ring substitution gives at least 0.086. This is the asymmetry an earlier element-free matcher measured and reported as its least convincing result, four cross-element pairings across six molecule pairs; here it follows from where the rewritten letter sits relative to the cycle rather than from a corpus. (C) The cyclic power profiles themselves, drawn as step functions around the six positions of the ring. Benzene is flat; each substitution raises one position, and it is the resulting dispersion—not the pattern of raised positions—that u reads. (D) A second registered expectation, also refuted. We predicted Δu would fall at three symmetric substitutions, since a symmetric trisubstitution restores the rotational symmetry of the substitution pattern (grey, dashed). It does not: the measured values rise monotonically $0.086 \rightarrow 0.105 \rightarrow 0.107$, because u measures dispersion of the profile’s values and symmetry of the pattern is not sameness of the values. An invariant reading the symmetry group of the substitution would behave as predicted; u is not that invariant, and we record the gap rather than redefine the quantity after seeing the result.

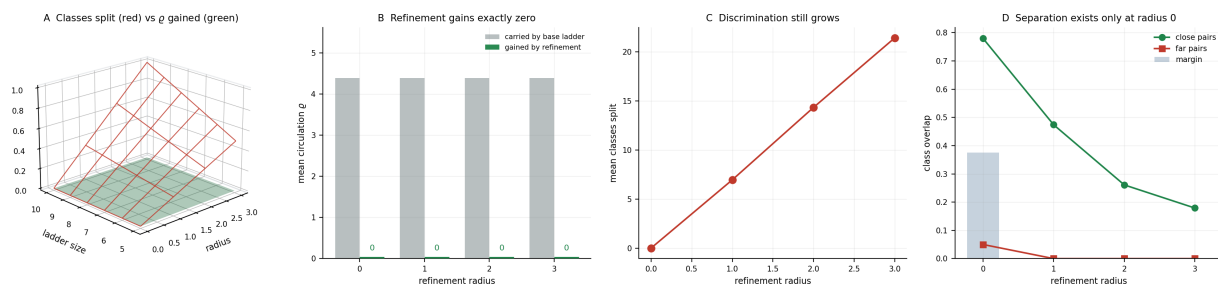


Figure 6: **Why a coarser comparison outperforms a finer one: refinement appends rungs that split classes while depositing no circulation.** (A) Two surfaces over refinement radius and ladder size. The flat green plane at zero is the circulation gained by refinement; the red wireframe rising above it is the number of label classes split. The gap between them is the whole of the effect. (B) Mean circulation gained per radius, which is zero to machine precision at every radius. Refinement appends step-pairs already traversed, and such steps are inert. (C) Mean classes split per radius over the same ladders, growing monotonically. The appended rungs do discriminate—they simply discriminate without determining anything new. (D) The measurement this explains, reproduced from the earlier element-free matcher: mean class overlap for bioisosteric (close) and unrelated (far) pairs against refinement radius, with the separation margin as bars. The margin is +0.375 at radius 0 and exactly 0.000 at radii 1, 2 and 3. Panels (B) and (C) supply a mechanism for panel (D); they do not re-measure it, and the text marks this category as a consistency check rather than as independent confirmation.

Table 1: Closed-ladder invariants for eight rings. Cyclohexane is perfectly uniform and not aromatic, refuting the uniformity formulation of aromaticity; circulation per rung separates the two classes with margin +0.293.

Ring	n	ϱ	ϱ/n	u	class
benzene	6	4.159	0.693	1.000	aromatic
naphthalene	10	6.931	0.693	1.000	aromatic
pyridine	6	4.433	0.739	0.914	aromatic
pyrimidine	6	4.708	0.785	0.895	aromatic
pyrazine	6	4.708	0.785	0.895	aromatic
cyclohexane	6	2.403	0.400	1.000	saturated
cyclopentane	5	2.002	0.400	1.000	saturated
cyclobutane	4	1.427	0.357	1.000	saturated

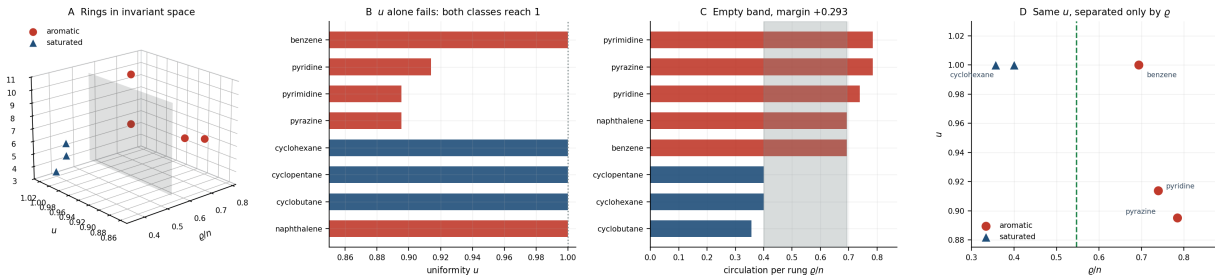


Figure 7: **A registered formulation of aromaticity, refuted by cyclohexane, and the corrected two-invariant statement.** (A) Eight rings in $(\varrho/n, u, n)$ space, aromatic in red and saturated in blue, with the separating plane at the midpoint of the circulation margin. The classes separate along the circulation axis and not along the uniformity axis. (B) Uniformity alone, sorted by ring. We predicted before running that a ring is aromatic exactly when its power profile is rotation invariant, i.e. when $u = 1$. Cyclohexane, cyclopentane and cyclobutane all attain $u = 1.000$ exactly and none is aromatic, so the prediction is refuted by the second-most familiar ring in chemistry. The panel is included because it is the refutation and not despite it. (C) Circulation per rung, same ordering. The shaded band is the empty interval between the largest saturated value (0.400) and the smallest aromatic one (0.693), a margin of +0.293; no ring lies inside it. Circulation is what separates the classes. (D) The two coordinates together. Benzene and cyclohexane sit at the same height ($u = 1.000$) and are separated only horizontally; benzene and pyridine sit at the same side of the threshold and are separated only vertically. Neither invariant classifies alone and the pair does, which is the corrected statement.

Remark 8.2 (What the refutation teaches). The failed formulation confused two different symmetries. Aromaticity is a property of *how much* each ring commitment closes—a magnitude, captured by ϱ —while uniformity is a property of *how evenly* the closures are distributed around the cycle. A saturated ring is perfectly even and weakly closing; a substituted aromatic is strongly closing and unevenly so. Only reading both coordinates separates the four cases, and we would not have found this had we not stated the single-invariant claim in advance and then measured it.

Remark 8.3 (Relation to the standard account). We make no claim to improve on the electronic account of aromaticity (Hückel, 1931; Schleyer, 2001) or on geometric indices derived from bond length equalisation (Krygowski et al., 2014). Theorem 8.1 is a statement about the invariants of a closed ladder on a set of profiles we assigned; the profiles are inputs, not measurements, and Section 11 records this as the principal limitation of the chemical part of this paper.

9 Validation

9.1 Method

Expectations were written into the experiment source before it ran and are reported below with their measured outcomes, including the two cases where the outcome contradicted the expectation. Every number quoted is read from the emitted JSON. Twenty-one checks are scored across seven categories; all twenty-one pass, but two of them pass by *refuting* the formulation we registered, which is why the count alone is uninformative and we report the content instead.

9.2 Closed-ladder invariants

Over 2000 random cycles, rotation deviation is 3.6×10^{-15} for ϱ and 4.4×10^{-16} for u ; the all-inert cycle has $\varrho = 0$ exactly. The identity $\varrho = -\log(1 - \pi(L))$ holds over 500 linear ladders, confirming the redundancy stated in Theorem 4.6 rather than concealing it. Of 4000 pairs constructed to share a composite power exactly, 2755 differ in u , which is the measurement establishing that u carries information composite power does not.

9.3 Substitution

Table 2: Change in uniformity under substitution. Position, not count, is what the prediction concerns; the monotonicity in the last column was predicted *not* to hold and does (Theorem 9.1).

Substitution	ring letters	u	Δu
peripheral (phenol/aniline)	0	1.000	0.000
one ring letter (pyridine)	1	0.914	0.086
two ring letters (pyrimidine)	2	0.895	0.105
two ring letters (pyrazine)	2	0.895	0.105
three ring letters (triazine)	3	0.893	0.107

The positional prediction holds with separation 0.086: peripheral substitution leaves u exactly unchanged and every ring substitution lowers it.

Remark 9.1 (A second registered expectation refuted). We predicted u would be *non-monotone* in the number of ring letters changed, reasoning that a symmetric trisubstitution restores the rotational symmetry of the substitution pattern and should return u toward its unsubstituted value. It does not: Δu rises $0.086 \rightarrow 0.105 \rightarrow 0.107$ and is monotone. The reason is that u as defined in Equation (2) measures dispersion of the profile’s *values* about their mean, and a symmetric substitution does not reduce that dispersion—symmetry of the pattern is not sameness of the values. An invariant sensitive to the symmetry group of the substitution pattern would

behave as we first expected; u is not that invariant, and we record the gap rather than redefine the quantity after seeing the result.

9.4 Inertness and repetition

Over 2061 steps of randomly generated histories on a seven-state set, with Δ computed as an explicit accumulating union: 0 violations of the biconditional in either direction, with 1716 fresh non-repetitions and 345 inert repetitions. Both branches are populated, so the biconditional is not satisfied vacuously by a run in which no step repeats—a control without which the result would be uninformative.

9.5 Refinement

Appended repetitions leave ϱ unchanged at every radius (mean gain 0 to machine precision) while the number of split classes grows monotonically with radius. Theorem 7.8 states what this does and does not establish.

9.6 Sensitivity

Over 5000 random ladders: the analytic derivative matches the numerical one to 1.6×10^{-10} ; the additive argmax is the highest-power rung in 5000/5000 cases, reproducing the prior claim; the proportional sensitivity is flat in 5000/5000 cases with maximum spread 1.1×10^{-16} .

9.7 Elimination

Over 3000 trials, two carriers realising the same contact sequence agree on the readout in every case. The two controls establish that this is informative: genuinely different sequences separate in $> 95\%$ of trials, and a carrier-dependent (non-local) contact effect separates in $> 95\%$. Without the second control the elimination result would be consistent with a readout that ignores its input.

Remark 9.2 (The first row of the elimination test is not evidence). That two carriers realising the same sequence agree is true by construction once a readout is defined to consult the terminal state alone. We report it as a consistency check. The load-bearing rows are the controls.

10 Relation to existing accounts

Metabolic control analysis. Control coefficients (Heinrich and Rapoport, 1974; Kacser and Burns, 1973) quantify the response of steady-state flux to enzyme concentration. Theorem 6.1 is a response of composite power to rung power, which is a different quantity, and Theorem 6.3 shows the comparison is further complicated by parametrisation. We flag the divergence rather than smooth it.

Amdahl’s law. Amdahl (1967) has the same multiplicative-complement structure as Theorem 3.4. That a result about parallel computation and one about chains of commitments share a form is what one should expect if neither depends on what the steps are.

Canonical forms. InChI (Heller et al., 2015) and canonical labelling (Morgan, 1965) determine a structure. A typed word does not: Theorem 4.7 already shows (ϱ, u) underdetermines a closed ladder. The comparison is between different tasks.

Fingerprints and refinement. Circular fingerprints (Rogers and Hahn, 2010) and colour refinement (Arvind et al., 2015; Weisfeiler and Leman, 1968) refine as far as possible to distinguish. Theorem 5.3 explains why the opposite is wanted here: past the tolerance point, refinement appends inert rungs.

Electron transport. We take nothing from the mechanistic literature (Marcus and Sutin, 1985; Mitchell, 1961; Moser et al., 1992) and make no claim about it; the transport chain enters this paper only as the construction whose abandonment produced the argument (Section 1).

11 Limitations

- L1. The ring profiles are inputs, not measurements.** Every number in Table 1 follows from profiles we assigned. The separation of aromatic from saturated rings is therefore a statement about the invariants given those profiles, and is not evidence that any real ring has them. Deriving profiles from structure is the work this paper does not do.
- L2. ρ is redundant for linear ladders.** Theorem 4.6. It earns its place only on cycles.
- L3. u is the wrong invariant for pattern symmetry.** Theorem 9.1. It reads dispersion of values, and an invariant reading the symmetry group of a substitution pattern would behave differently. We have not constructed one.
- L4. The refinement result is a consistency check.** Theorem 7.8. It does not re-measure the radius sweep.
- L5. Theorem 3.5 is relative to its observable class,** which is a stipulation. The result that is not a stipulation is Theorem 2.3, and it carries hypotheses that fail in a regime real systems occupy (Theorem 2.5).
- L6. Which sensitivity parametrisation is physical is not settled.** Theorem 6.3 shows the counter-intuitive direction depends on it, and we do not determine which applies to any real system.
- L7. No endpoint validation.** Nothing here is tested against spectra, activity data, binding data, or any retrieval benchmark. The measurements are internal to the formalism.
- L8. Powers are not derived.** As in both prior developments, a power is measured rather than predicted. The framework says what follows from powers, not what determines them.

12 Conclusion

A process whose carrier has been deleted is a ladder, and two prior developments established that for linear ladders the numbers are the whole description. Molecules are not linear ladders; they contain cycles, and a cycle has no target. We supplied two invariants for the closed case, and were explicit that one of them, the circulation, is a reparametrisation of composite power in the linear case and earns its place only by surviving the passage to a cycle.

Two formulations we registered in advance were refuted by our own measurements. Aromaticity is not rotational invariance of the power profile: cyclohexane is perfectly uniform and not aromatic, and the corrected statement needs circulation per rung to separate aromatic from saturated (+0.293) and uniformity to separate substituted from unsubstituted, with neither sufficient alone. And uniformity is not non-monotone in the number of ring substitutions, because it reads dispersion of values rather than symmetry of pattern.

We also narrowed the sharpest claim of both prior papers. Control lies at the strongest rung under additive improvement—which we reproduce, 5000/5000—and is flat under proportional improvement, spread 1.1×10^{-16} . The check both papers ran cannot distinguish these, because it compares two computations of the same additive derivative.

What the chemical application buys is narrower than the framing might suggest and is worth stating exactly. Because a typed word is ordered, a correspondence between molecules is

connected by construction, which removes the first two stated limitations of an earlier element-free matcher. And because circulation and uniformity are functions of a cycle’s profile alone, the positional asymmetry that matcher measured without explaining—peripheral substitution preserving structural class, ring substitution degrading it—is predicted rather than fitted: 0.000 against ≥ 0.086 . What is not bought is the profiles themselves, which we assigned. Until those are derived from the shell arithmetic that already yields valence and geometry, the chemical claims of this paper are statements about a formalism and not about chemistry.

Reproducibility

The experiment source records its expectations before the measured values and writes every reported quantity to JSON. Retained in the source with their status marked are the refuted uniformity formulation of aromaticity (Section 8), the refuted monotonicity expectation (Theorem 9.1), the redundancy of ρ in the linear case (Theorem 4.6), the tautological first row of the elimination test (Theorem 9.2), and the limited scope of the refinement category (Theorem 7.8). The only stochastic components seed explicitly.

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